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SWEN90010 - High Integrity Systems Engineering

High Integrity Systems, Safety

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INTRODUCTION To SAFETY ENGINEERING

Outcomes

What is “safety”?

What does it mean for software?

How do we get it? (safety engineering)

Safety

?

Doesn't cause unacceptable harm to
the environment or people

Safety

e.g. we accept some risk of planes crashing

Doesn't cause **unacceptable** harm to
the environment or people

How we define and quantify
unacceptability is a big
part of safety engineering.

Software



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Can software harm anyone on its own?

```
except socket.error, (errno, strerror):  
    print "ncfiles: Socket error (%s) for host %s (%s)" % (errno, strerror, host)  
    continue  
    print "ncfiles: urllib2 error (%s)" % msg  
    continue  
for h3 in page.findAll("h3"):  
    value = (h3.contents[0])  
    if value != "Afdeling":  
        print >> txt, value  
        import codecs  
        f = codecs.open("alle.txt", "r", encoding="utf-8")  
        text = f.read()  
        f.close()  
        # open the file again for writing  
        f = codecs.open("alle.txt", "w", encoding="utf-8")  
        f.write(value+"\n")  
        # write the original contents  
        f.write(text)  
        f.close()
```

No

Only when in the right **context**

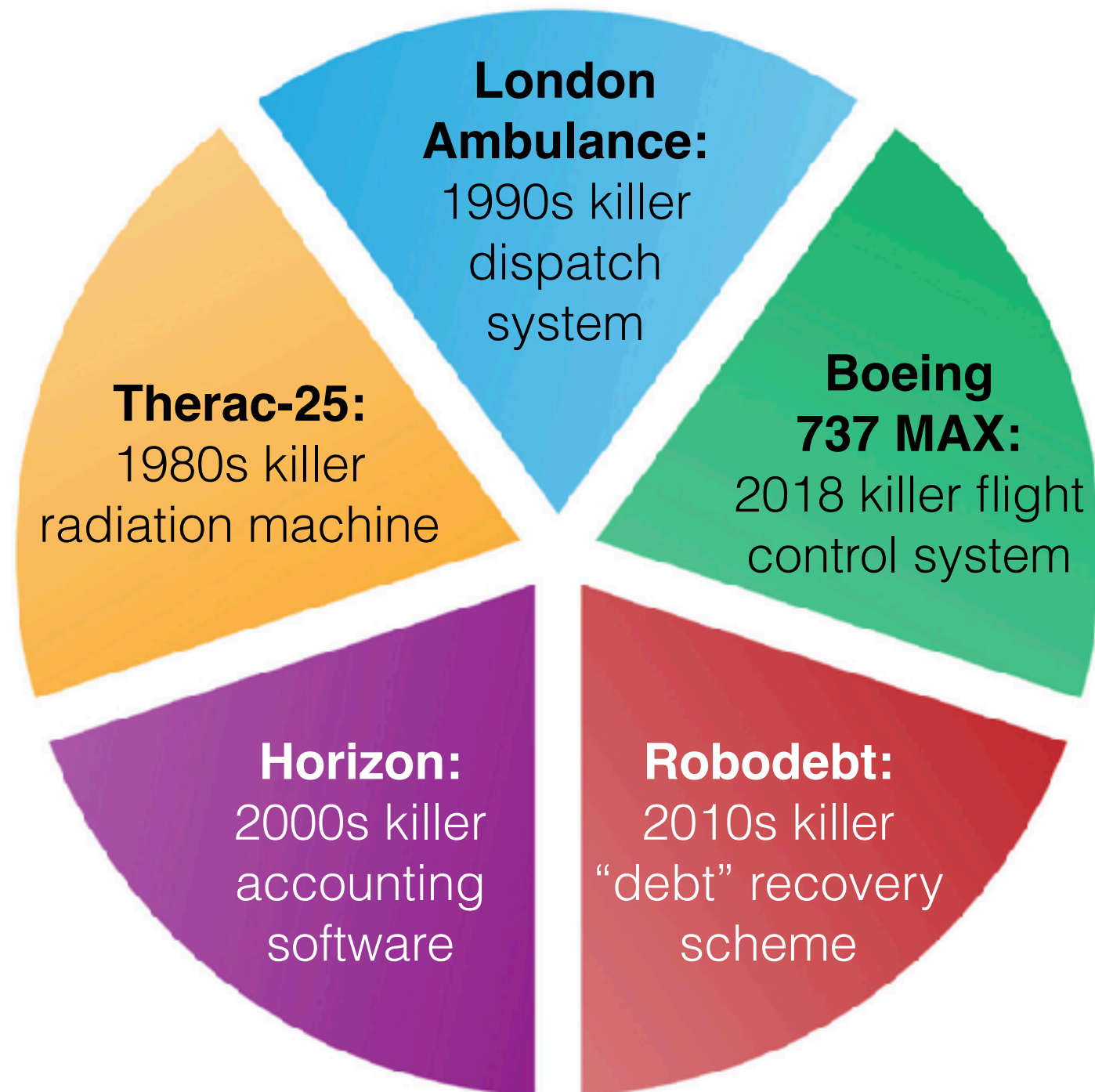
Only as part of a **larger system**

Software engineering is **always** about the larger system,
but even more so for safety.

Killer Software Roulette



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Live Results



[https://www.polleverywhere.com/ranking_polls/
3CnYqTSxglOLu8DqCd4aJ?preview=true&controls=none](https://www.polleverywhere.com/ranking_polls/3CnYqTSxglOLu8DqCd4aJ?preview=true&controls=none)



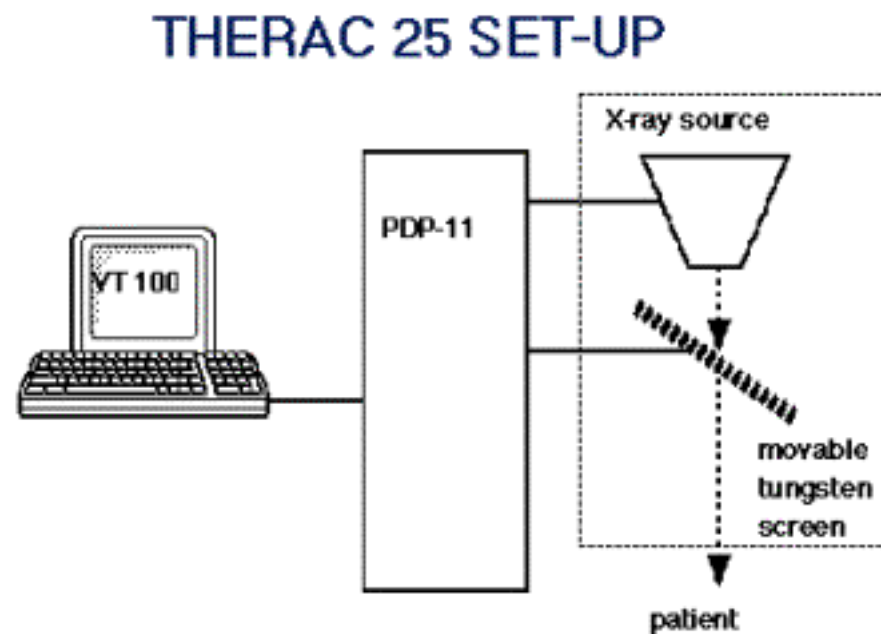
CASE STUDY: THERAC-25

Therac-25



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mid-1980s Radiation Therapy Machine



Therac-25 Harm

June 1985 - Jan 1987: 6 radiation overdoses,
multiple deaths.

March 21, 1986 (3rd incident): “Inside the treatment room Cox was hit with a powerful shock. He knew from previous treatments this was not supposed to happen. He tried to get up. Not seeing or hearing him because of the broken communications between the rooms, the technician pushed the "p" key, meaning "proceed." Cox was hit again. The treatment finally stopped when Cox stumbled to the door of the room and beat it with his fists.”

He was sent home but returned to the hospital a few weeks later ... diagnosed radiation overexposure. It later paralysed his left arm, both legs, his left vocal chord, and his diaphragm. He died nearly five months later.

Therac-25 Causes

Software errors

“Therac-25 typically issued up to four error messages a day”

“At the computer console she typed in the prescription data for an electron beam of 180 rads, then noticed she'd made an error by typing in command x (for x-ray treatments) instead of e (for electron). She ran the cursor up the screen to change the command x to e”



Poor Usability

Software Overconfidence



“On Therac-6 and 20 hardware lockout mechanisms do not allow the operator to do something dangerous”

“When it came time to make Therac-25, AECL decided to keep only the computer control. They not only refused manual controls, but also the hardware lock mechanisms.”

Manufacturer refused to believe machine was faulty initially, until more incidents.



CASE STUDY: LONDON AMBULANCE DISPATCH SYSTEM

26 October 1992

“Just a few hours [into its deployment] AVLS was unable to keep track of the ambulances and their statuses in the system. It began sending multiple units to some locations and no units to other locations. ... The system began to generate such a great quantity of exception messages ... that calls got lost. The problem was compounded when people called back additional times because the ambulances they were expecting did not arrive. ... The next day, the LAS switched back to a part-manual system, and shut down the computer system completely when it quit working altogether eight days later”

AVLS Harm

“There were as many as 46 deaths that would have been avoided had the requested ambulance arrived on time.”

“One heart attack patient waited six hours for an ambulance before her son took her to the hospital.”

“Another woman called the LAS every 30 minutes for almost three hours before an ambulance arrived. It was too late, as her husband had already died.”

“One ambulance crew arrived only to find that the patient had not only died, but his body had been taken away by a mortician.”

LAS Since

Further systems launched in 2005 and 2011,
both extremely unreliable initially.

Ambulance scheduling is not exceedingly
difficult.

Problem is that these systems weren't engineered
as if they were safety critical.



CASE STUDY: AIRBUS A320

Airbus A320

A computer network in a plane.

~150 ECUs, so a big distributed system.

First civilian fly-by-wire aircraft.

26 June, 1988

Airbus A320 first passenger flight
Low-speed flyover at Habsheim Air Show,
at 10 metres altitude



One woman and two children died.

What Happened

Official Report

Pilot flew too low, too slow.

Failed to see forest.

Cause: Pilot Error

Not (only)
pilot error:
system failure.

Captain's Report

Fly-by-wire system
prevented plane from
levelling at correct
altitude and climbing.

Forest not shown on
airport map given to pilots.

Expecting different
(longer) runway



CASE STUDY: BOEING 737 MAX

Summary

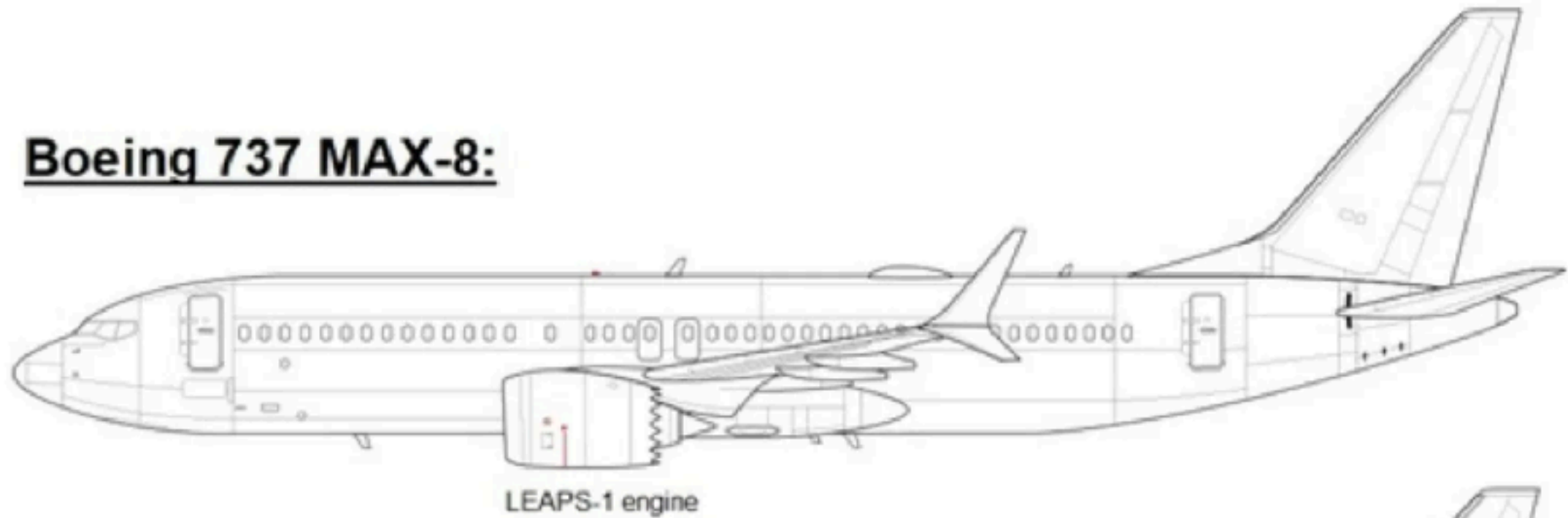
- 29 October 2018: Lion Air Flight 610 crashes
- 10 March 2019, Ethiopian Airlines Flight 302 crashes
- Both Boeing 737 MAX aircraft
- 346 people killed
- General consensus that major contributing factor was **MCAS** system (which is a new system added to 737 MAX series)

Why MCAS?

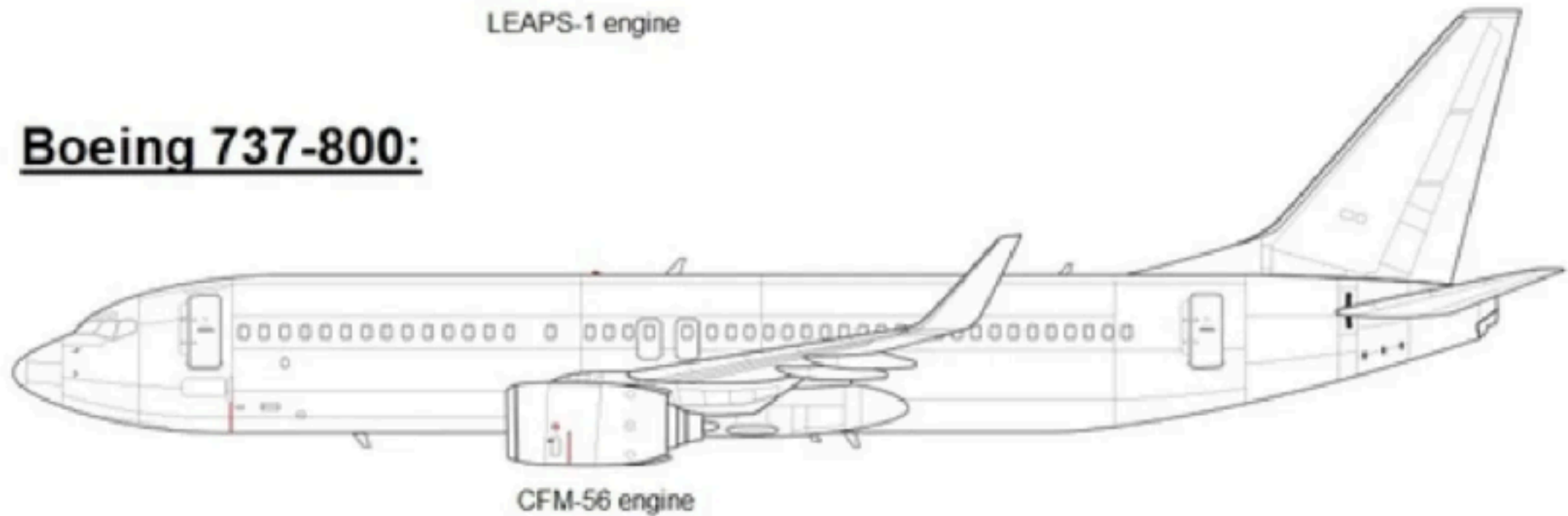


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Boeing 737 MAX-8:



Boeing 737-800:



MCAS Function



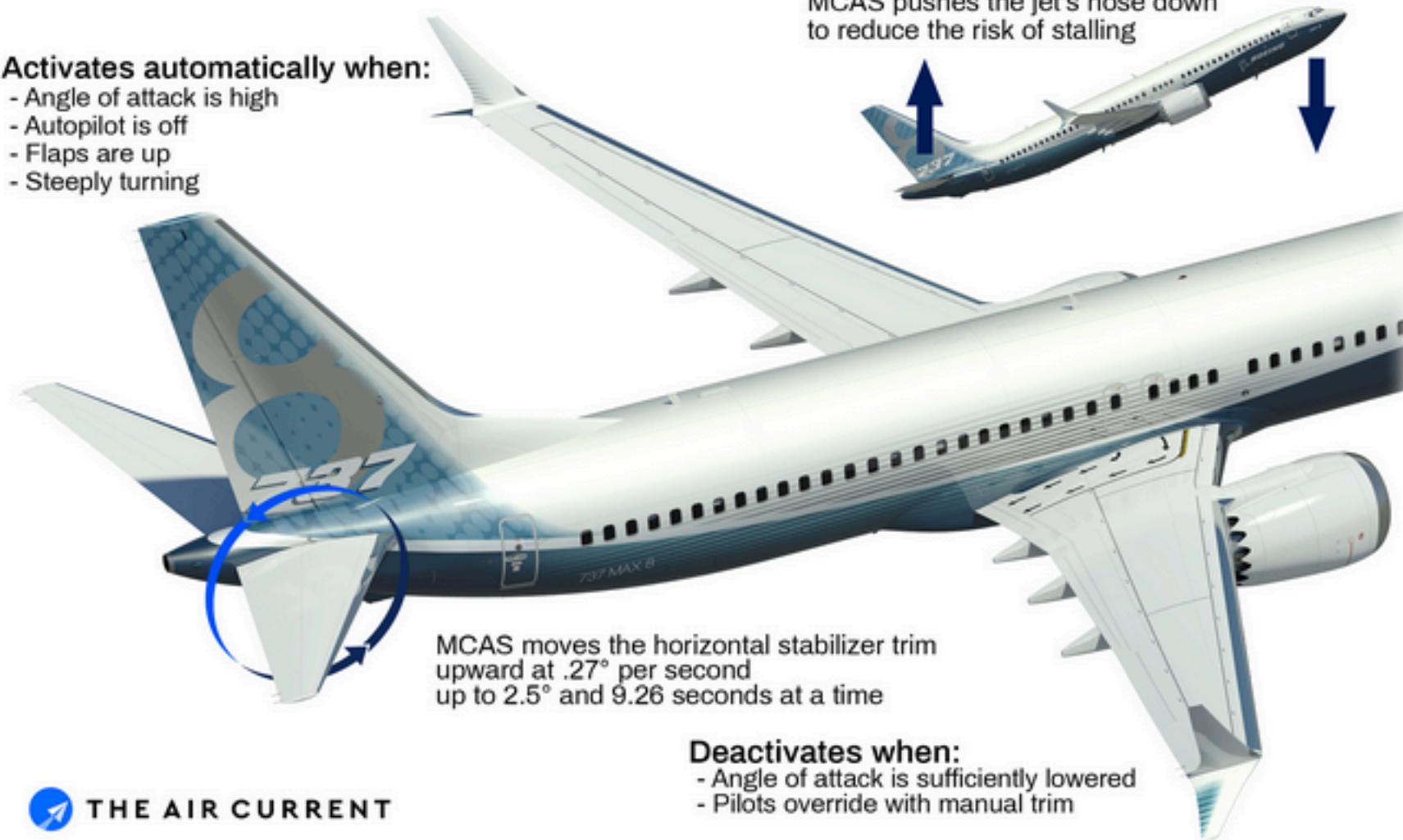
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Boeing 737 Max Maneuvering Characteristics Augmentation System

Activates automatically when:

- Angle of attack is high
- Autopilot is off
- Flaps are up
- Steeply turning

MCAS pushes the jet's nose down
to reduce the risk of stalling



MCAS moves the horizontal stabilizer trim
upward at $.27^\circ$ per second
up to 2.5° and 9.26 seconds at a time

Deactivates when:

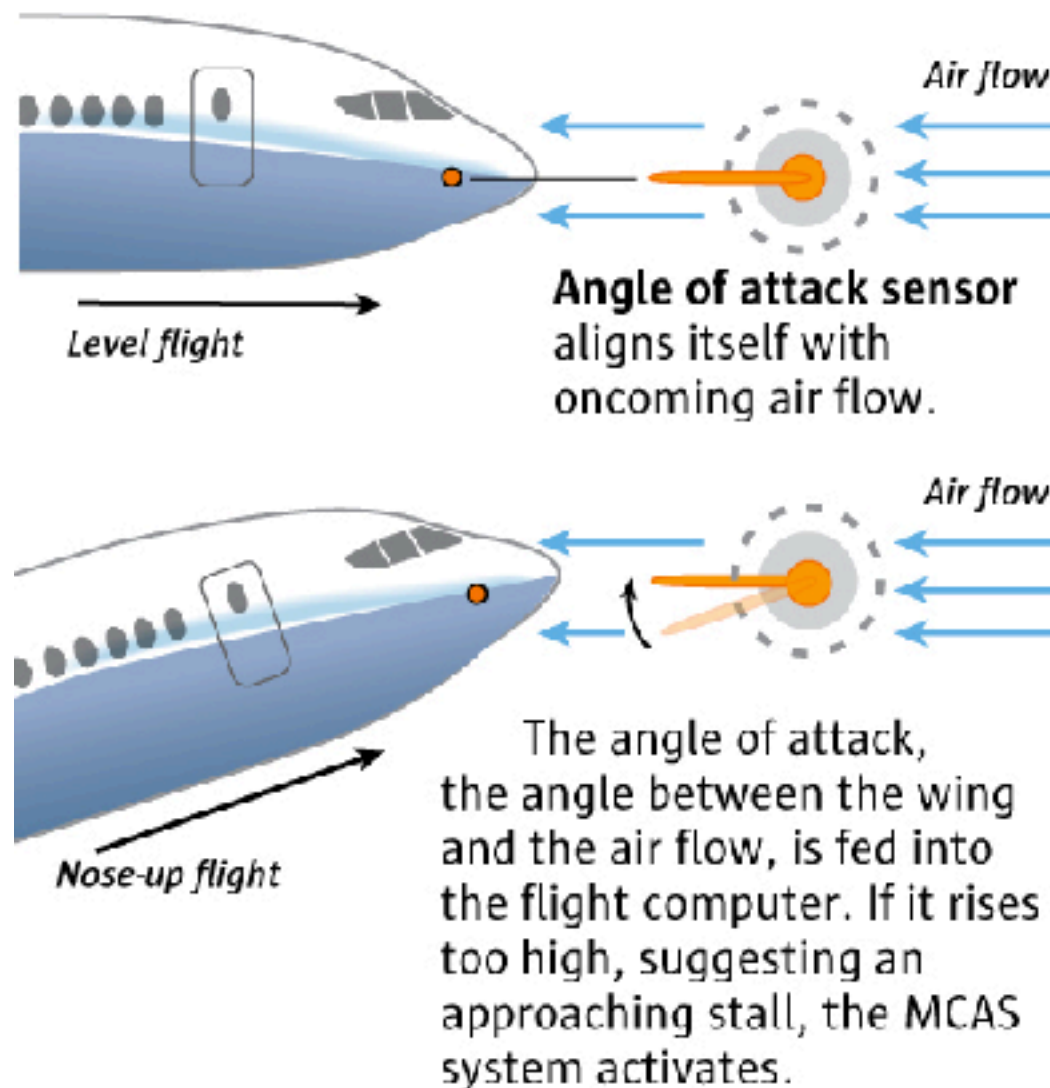
- Angle of attack is sufficiently lowered
- Pilots override with manual trim

 THE AIR CURRENT

Angle of Attack Sensors

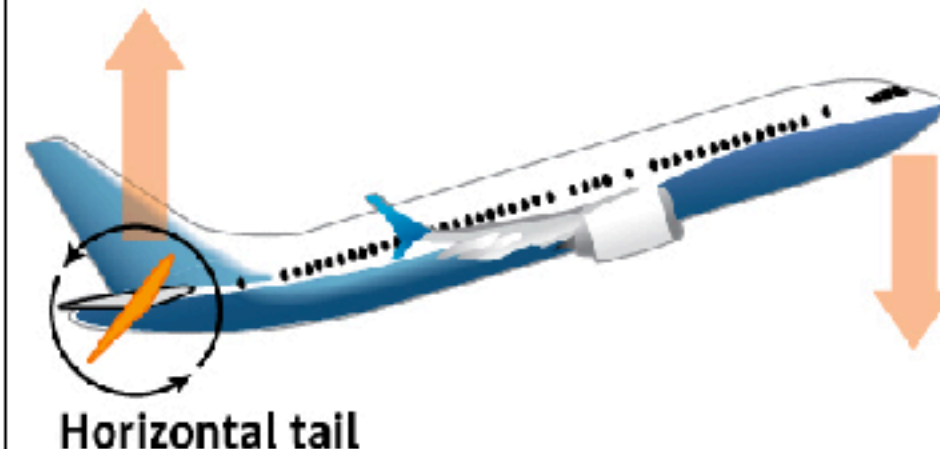


How the new MAX flight-control system operates to prevent a stall



MCAS (Maneuvering Characteristics Augmentation System)

The MCAS system automatically swivels the horizontal tail to move the nose down. In the Lion Air crash, the angle of attack sensor fed false information to the flight computer.



Sources: Boeing, FAA, Indonesia National Transportation Safety Committee, Leeham.net, and The Air Current.

Reporting by DOMINIC GATES,
Graphic by MARK NOWLIN / THE SEATTLE TIMES

Immediate Cause

MCAS repeatedly activated

Repeatedly pushed plane's nose down

Pilots were unable to correct its effects

Plane eventually crashed into the ground

MCAS repeated activated believed to
be caused by receiving faulty AoA
readings

Contributing Factors

Lack of redundancy: MCAS took input from only **one** AoA sensor (despite the 737 MAX having two AoA sensors)

Lack of human override: Pilots repeatedly tried to bring plane nose up by moving the control column. But MCAS was designed so that it couldn't be overridden this way.

Lack of Documentation/Training: MCAS not described in manual or training — apparently deliberate choice to maximise similarity to prior 737s to minimise total cost of ownership

Culture: Economics vs Safety



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MCAS resulted from an **economic incentive** to make the 737 MAX as economically attractive to potential buyers as possible

(Bigger engines = **more efficient plane**;
but necessitated MCAS to maximise
similarity to prior 737s)

(Maximising similarity = **minimal retraining**;
but meant pilots were unaware of MCAS)

(Using only 1 AoA sensor = **no need to re-certify**;
but meant a single point of failure)

Consequences

346 deaths

Indefinite grounding of all 737 MAXs
pending investigation and re-certification

Projected cost: 18.4 billion USD

CEO removed

2019: airlines cancelled 183 orders
(at \$100M per plane = 18.3 billion USD)

**Cheapest way to build something is to
build it right the first time**

Culture and Ethics

Engineers have a responsibility to act ethically

In cultures that (implicitly) prioritise **economics** over **safety** or other concerns. this may require challenging that culture or refusing to follow management directives

Remember: The first employee (James Liang) of Volkswagen sent to prison over its emissions cheating system was a software engineer



HORIZON IT SYSTEM (UK POST OFFICE)

Horizon IT System (UK)

- Accounting software used by British Post Office
- In the UK, most post offices are run like franchises, by independent business people called “sub postmasters”
- Part of Horizon’s job was to track accounting shortfalls of subpostmasters
- Horizon was faulty, resulting in false shortfalls for **thousands** of subpostmasters
- 1999-2015: 900+ convictions for theft, fraud, false accounting
 - Despite prior to 1999 having very low rates of fraud etc.



Horizon Harms

- Post Office carried many of the prosecutions (~700)
- Besides criminal convictions, others lost their jobs or had to pay the shortfalls themselves
- Bankruptcies, divorces, suicides
- **“one of the greatest miscarriages of justice in our nation’s history” - Prime Minister Rishi Sunk (2014)**

Want to know more?



<https://7plus.com.au/mr-bates-vs-the-post-office>



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ROBODEBT



THE FLAWED ALGORITHM AT THE HEART OF ROBODEBT

Robodebt teaches us that even simple automated decision-making systems come with the biases of the people, systems and policies that conceive them

By Associate Professor Toby Murray, Dr Marc Cheong and Professor Jeannie Paterson, University of Melbourne



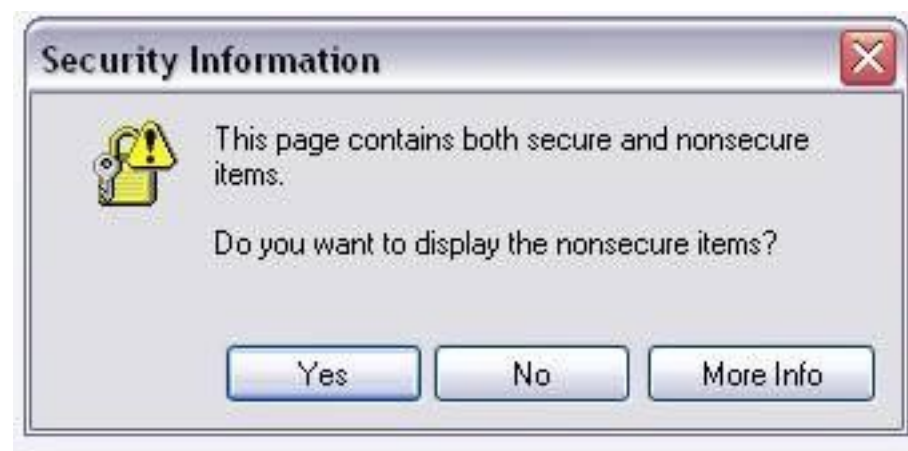
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SAFETY ENGINEERING

Importance of The System

Safety cannot be talked about
without considering the entire **system**

e.g. a pilot might fail to notice a signal,
and be blamed (“pilot error”),
but blame is tricky to assign if the
signal was placed in the cockpit
at a location that was hard to see.



c.f. early SSL/TLS
warning dialogs

The System

Software

Hardware

Operating Procedures
(this is where people, users, come in.)

The **context** in which the
system sits.

Safety

In light of the above, a more precise definition:

**software and hardware used under correct
operating conditions don't cause unacceptable
harm to people or environment.**



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WHO HAS NOT FOUND A PARTNER YET?